Banking Credit Risk Calculator: A Basel III Compliant, ML-Powered Credit Risk Platform

PD Modelling · Policy Governance · RM Decision Support · Regulatory & Financial Reporting

Credit Risk & Banking Intelligence Platform

Banking Credit Risk Calculator — Internal Solution Overview

July 2026

Outline

What this platform is

A single Flask-based application that takes a bank from **raw borrower data** to a **governed credit decision** to **regulator-ready reporting** — covering origination, ML risk scoring, deterministic policy, human-in-the-loop approval, Basel III capital/liquidity reporting, and financial statements, for one global banking group.

Why it matters

- Replaces spreadsheet-based, disconnected credit assessment with a single traceable pipeline: model → policy → human decision → report
- Every recommendation is **explainable** (SHAP-driven attribution) and every RM decision is **hash-chained** for audit
- Basel III capital/liquidity ratios and financial statements are derived from the **same underlying loan ledger** — not reconciled separately

Scale (grounded ledger)

- 9 real-world banks across 6 countries (India, USA, UK, Singapore, UAE, Australia)
- Group → Region → Country → Bank hierarchy
- All capital/liquidity/NPA/P&L figures roll up from per-loan records

Six Departments, One Platform

Department	What it does	Route
Credit Risk	ML PD, LGD, RWA, EL, pricing — single-borrower calculator	`/`
Banking Operations	Multi-bank dashboards, customer/account/loan ledger	`/operations/`
Regulatory Reporting	Basel III/RBI capital (CRAR, LCR, NSFR) per bank & client	`/regulatory/`
Relationship Mgmt	Machine recommendation → RM accept/reject → PDF report	`/relationship/`
Financial Reporting	Balance sheet, P&L, key ratios, Pillar 3 — any roll-up scope	`/financials/`
Global Reference Data	Countries, macro indicators, sovereign ratings	`/reference/`

One origination channel, six consuming views

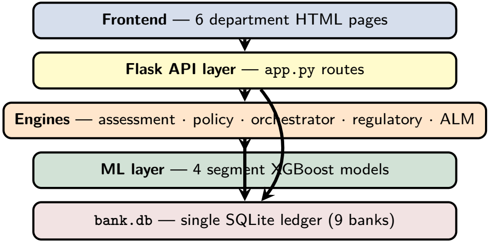
A borrower is captured **once** on the Credit Risk calculator. Every other department (Operations, Regulatory, RM, Financials) reads from the same ledger — there is no duplicate data entry and no reconciliation step between departments.

Architecture & Technology Stack

Layer	Technology
Web server	Flask 3.x + gunicorn
ML model	XGBoost — 4 segment models
Scheduler	APScheduler (background)
Database	SQLite `bank.db` (raw SQL)
Frontend	Vanilla HTML5 / CSS3 / JS
Deployment	GCP App Engine (Py 3.10)

Key backend modules

- `assessment_engine.py` — full borrower findings
- `policy_engine.py` — deterministic rules
- `decision_orchestrator.py` — Machine Recommendation
- `shap_explainer.py` — feature attribution
- `regulatory_engine.py` — Basel III / RBI
- `alm_engine.py` — asset-liability management



Probability of Default (PD) — Segment-Specific XGBoost

A single aggregate model under-serves heterogeneous portfolios — a Corporate borrower and a Retail-Mortgage borrower default for different reasons. The platform trains **one XGBoost classifier per Basel exposure class**, each promoted independently.

Exposure Class	AUC-ROC	Accuracy	Recall	SA Risk Weight
CORPORATE	0.754	60.9%	73.6%	100%
SME	0.764	63.7%	79.4%	85%
RETAIL_MORTGAGES	0.758	63.0%	76.5%	35%
RETAIL_OTHER	0.753	65.4%	78.8%	75%

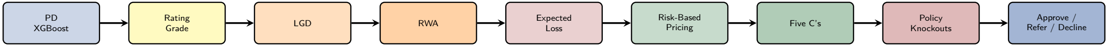
Feature schema (27 inputs)

- Core financial ratios: `de_ratio`, `interest_coverage`, `profitability`, `liquidity_ratio`
- 16 borrower / bureau attributes (age, income, CIBIL, FOIR, employment, delinquency history, EMI-miss ratio, income volatility)
- 4 country macro indicators + 3 trend features

Governance

- `model_feature_frame()` aligns any partial input to the full schema (neutral defaults for missing fields)
- Promotion requires R^2 improvement over the active model; prior model is retained as `pd_model_backup.pkl`
- Full run history & 5-fold CV logged per training run

From Raw Data to a Priced Decision: the Assessment Pipeline



One findings object

`assessment_engine.py` produces a single JSON object with every stage above — consumed by both `report-underwriter.html` (internal) and `report-applicant.html` (adverse-action letter).

Explainability

`shap_explainer.py` attaches SHAP-based feature attribution and counterfactual recourse — *"what would need to change for this applicant to qualify."*

PD is converted into a 10-grade internal scale aligned to Standardised-Approach external rating equivalents — consistent scoring language across ML output and regulatory reporting.

Grade	PD Band	Label	Inv. Grade	Decision
AAA	0.00% – 0.50%	Exceptional	Yes	**APPROVE**
AA	0.50% – 1.00%	Very Strong	Yes	**APPROVE**
A	1.00% – 2.00%	Strong	Yes	**APPROVE**
BBB	2.00% – 4.00%	Adequate	Yes	**APPROVE**
BB	4.00% – 7.00%	Speculative	No	**REFER**
B	7.00% – 12.00%	Vulnerable	No	**REFER**
CCC	12.00% – 20.00%	Distressed	No	**DECLINE**
CC	20.00% – 35.00%	Highly Distressed	No	**DECLINE**
C	35.00% – 50.00%	Near Default	No	**DECLINE**
D	50.00% – 100%	Default	No	**DECLINE**

Deterministic Credit Policy — Independent of the Model

Why a separate policy engine?

The model produces a **risk estimate**; `policy_engine.py` applies the bank's **eligibility rules and risk-appetite cutoffs**. A decline can be proven to a regulator as policy-driven, not a black box — and policy can be re-versioned (`credit-policy-v1.0`) without ever retraining the model.

Rule	Threshold	Outcome
Sanctions / AML hit	any match	**DECLINE** (hard gate)
KYC incomplete	status $\neq$ Verified	**DECLINE**
PEP flag	–	**REFER** + Enhanced DD
FOIR (debt-service)	$\geq$45% refer / $\geq$55% decline	REFER / DECLINE
Minimum income by product	Home Rs. 5L, Business Rs. 6L, Personal Rs. 3L	REFER if below
LTV cap (secured products)	Home 80% / Vehicle 85% / Business 70%	REFER if breached
Age	$<$21 decline, maturity age $>$70 refer	DECLINE / REFER
Single-borrower exposure cap	Rs. 2 crore	REFER + committee flag

Indicative Rate = Base Rate + EL Spread + Capital Charge + Operating Margin

Component	Default
Base rate (RBI repo)	6.50%
Min. capital ratio	8% (Basel III Pillar 1)
Target ROE	15%
Operating margin	150 bps
Floor / Ceiling	7.0% / 24.0%

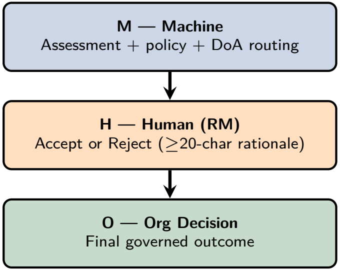
- **EL Spread** $=$ PD $\times$ LGD, expressed in bps — the minimum margin to break even on expected credit losses
- **Capital Charge** $=$ (RWA $\times$ 8%) $\times$ target ROE, expressed as bps over EAD — cost of holding regulatory capital
- Rate above the 24% ceiling is flagged **"unviable"** — the platform will not silently over-price a bad risk

`decision_orchestrator.py` composes model findings + policy + confidence/OOD routing + knife-edge sensitivity into a single **Machine Recommendation (M)** — then decides who is authorised to act on it.

Control Level	Trigger	Meaning
`RM_SINGLE`	Exposure $\leq$ Rs. 20L, high confidence	RM may finalise alone
`FOUR_EYES`	Exposure $>$ Rs. 20L, or LOW confidence, or knife-edge PD, or REFER	Requires credit-officer approval
`COMMITTEE`	Exposure $>$ Rs. 1 crore, or committee policy flag	Risk Committee only
`AUTO_DECLINE`	Hard decline / financial-crime escalation	RM cannot approve

Knife-edge detection

If the point-estimate PD sits within $\pm$20% of a decision cutoff (REFER at 12%, DECLINE at 50%), the recommendation is flagged **sensitive** — a small input change could flip the outcome — forcing four-eyes review even when the exposure alone would allow RM-single authority.



- RM has **two actions only** — accept or reject; both finalise immediately (no partial states)
- Every event is written to a **hash-chained** `rm_case_events` table — tamper-evident provenance from M to O
- Outcomes feed `rm_outcomes` for HITL (human-in-the-loop) learning via `/relationship/api/insights`
- **Single origination channel** — data is captured once on the calculator's borrower form; the RM's "+ New Customer" simply navigates back to it

Case Lifecycle: Onboarding to Signed-Off Report

1. Assess	2. Refer to RM	3. RM Decision	4. Generate Report	5. Deliver PDF
`POST /api/assess-borrower`	`POST /relationship/api/cases`	`POST .../cases/id/action`	`POST .../cases/id/report`	`GET .../reports/id/ver/pdf`

→ each step hands off to the next; the case is never re-entered at an earlier stage.

Underwriter report — 6 sections

1 Risk Assessment · 2 Five C's Credit Assessment (2b Macro Regime Score) · 3 Peer Comparison · 4 Improvement Pathways · 5 AIRB Regulatory Details · 6 Credit Decision.

PDF generation & versioning

`report_generator.py`: matplotlib charts → LaTeX → pdflatex. Every report is versioned under `data/case_reports/<case_id>/<version>/` — regenerating a report never overwrites a prior version.

Banking Operations (`/operations/`)

- Multi-bank dashboard — 9 banks, live customer/account/loan ledger
- Loan portfolio by status (Active, NPA, Defaulted, Written-Off, Closed)
- Per-customer 360° view: accounts, loans, transactions
- NPA resolution workflow; live DB schema viewer for admins

Regulatory Reporting (`/regulatory/`)

- Basel III / RBI: CRAR $\geq$11.5%, Tier-1 $\geq$9.5%, CET1 $\geq$8%
- Liquidity: LCR $\geq$100%, NSFR $\geq$100%, CRR $\geq$4.5%, SLR $\geq$18%
- Per-client RWA exposures rolled up to bank-level capital reports
- Daily batch (APScheduler, 01:00) + on-demand `run-batch`

Same ledger, no reconciliation

Both departments read the *same* `bank.db` loan/account records that the Credit Risk calculator writes to — capital ratios and NPA figures are always consistent with the underlying originated exposures.

Financial Reporting & Global Reference Data

Financial Reporting (`/financials/`)

- Balance Sheet, P&L, Key Ratios, Pillar 3 disclosures
- One generic `consolidate()` function produces bank / region / country / group roll-ups from the same code path
- Versioned PDF annual reports per scope
- Foreign banks use a 75% average risk weight where loan-level detail is unavailable

Global Reference Data (`/reference/`)

- `countries` — ISO-3, region, Basel minimums, sovereign rating
- `country_macro` — GDP, CPI, policy rate, FX (2 periods/country)
- 7 countries seeded; feeds the ML model's macro features and the onboarding form's country selector

Global hierarchy

Group → Region → Country → Bank. 9 real-world banks grounded in FY2025 annual reports — India (HDFC, ICICI, Bank of Baroda, PNB), USA (JPMorgan Chase), UK (Barclays), Singapore (DBS), UAE (Emirates NBD), Australia (Commonwealth Bank) — all foreign ledgers denominated in group reporting currency (Rs.).

Four Exposure Segments, One Onboarding Form

The same `borrower-info.html` intake form serves every customer type — an **Exposure Class** dropdown routes the application to the matching PD model, risk weight, and downstream policy checks. The next four slides walk each segment end-to-end.

Exposure Class	Typical Customer	SA RW	PD AUC	Key Inputs
CORPORATE	Large corporate borrower	100%	0.754	DE ratio, interest coverage, profitability
SME	Small/medium enterprise	85%	0.764	Liquidity ratio, turnover, collateral
RETAIL_MORTGAGES	Home loan applicant	35%	0.758	Income, LTV, CIBIL, employment
RETAIL_OTHER	Personal/vehicle/education	75%	0.753	FOIR, CIBIL, dependents, tenure

Common intake fields (all segments)

Borrower ID/Name, Exposure Amount, Country, KYC status, sanctions screening, proposed EMI, existing obligations, sector, loan type (auto-sets maturity), collateral type/value, seniority, LTV. Retail applications add age, employment, CIBIL score, FOIR, dependents; Corporate/SME applications add the four financial ratios that feed the model directly.

Use Case 1 — CORPORATE: Large Working-Capital Facility

Scenario: A mid-size manufacturer requests a Rs. 15 crore working-capital facility.

1. **Onboard** — RM enters borrower financials: DE ratio, interest coverage, profitability margin, liquidity ratio; Exposure Class = `CORPORATE`
2. **Score** — CORPORATE XGBoost model returns PD; masterscale maps it to a grade (e.g. BBB, 3.1% PD → APPROVE band)
3. **Policy** — single-borrower cap (Rs. 2 crore) is exceeded by the requested Rs. 15 crore → `SINGLE_BORROWER_CAP` fires, `COMMITTEE_REQUIRED` flag set
4. **Route** — exposure > Rs. 1 crore → `COMMITTEE` control level; RM cannot finalise alone
5. **Decide & Report** — Risk Committee reviews the Machine Recommendation, accepts with conditions; underwriter PDF generated and versioned

Why this matters

Large corporate exposures automatically escalate past RM authority — the platform enforces committee governance *before* any human has a chance to approve outside policy.

Output

Rating grade, RWA/EL/capital charge, indicative pricing, Five C's narrative, committee-ready PDF with full audit trail.

Use Case 2 — SME: Business Expansion Loan

Scenario: A retail-chain SME requests a Rs. 35 lakh business loan against Rs. 50 lakh collateral.

1. **Onboard** — Exposure Class = `SME`; LTV = 35L/50L = 70% (exactly at the Business Loan policy cap of 70%)
2. **Score** — SME model (AUC 0.764) returns PD; grade BB → REFER band
3. **Policy** — LTV at the boundary passes; minimum-income gate for Business Loan (Rs. 6L) is checked against the SME's declared income
4. **Route** — BB grade + REFER model output → `FOUR_EYES` control level even though exposure (Rs. 35L) is above the Rs. 20L RM-single ceiling anyway
5. **Decide & Report** — Credit officer reviews SHAP attribution and counterfactual recourse ("what would move this to APPROVE"), accepts with conditions (reduce quantum or add collateral)

Why this matters

SME risk sits in the volatile middle band — the platform's knife-edge sensitivity check is exactly the safeguard this segment needs.

Output

Conditional approval with explicit remediation levers, indicative rate reflecting the 85% SA risk weight, applicant-facing adverse-action letter if declined.

Use Case 3 — RETAIL_MORTGAGES: Home Loan

Scenario: A salaried individual, age 34, requests a Rs. 45 lakh home loan against a Rs. 60 lakh property.

1. **Onboard** — Exposure Class = `RETAIL_MORTGAGES`; LTV = 45L/60L = 75% (within the 80% Home Loan cap); CIBIL, FOIR, employment tenure captured
2. **Score** — RETAIL_MORTGAGES model (AUC 0.758) returns PD; grade A → APPROVE band
3. **Policy** — income Rs. 8L > Rs. 5L Home Loan minimum, age 34 with 20-year tenor → maturity age 54 < 70 cap — all rules clear
4. **Route** — exposure Rs. 45L > Rs. 20L RM-single ceiling → `FOUR_EYES`, but high confidence and no knife-edge means fast-track review
5. **Decide & Report** — Credit officer accepts; borrower receives a plain-language applicant letter with indicative rate at the 35% mortgage risk weight

Why this matters

Mortgages carry the lowest SA risk weight (35%) — pricing and capital charge are materially lower than an equivalent unsecured personal loan, visible directly in the pricing breakdown.

Use Case 4 — RETAIL_OTHER: Personal Loan (Declined, then Recourse)

Scenario: An applicant with FOIR 58% requests a Rs. 4 lakh personal loan.

1. **Onboard** — Exposure Class = `RETAIL_OTHER`; FOIR computed from existing obligations + proposed EMI over monthly income
2. **Score** — RETAIL_OTHER model (AUC 0.753) returns PD; independently, FOIR 58% $\geq$ 55% `FOIR_EXCEEDS_LIMIT` $\rightarrow$ **DECLINE**
3. **Policy dominates** — `_compose()` takes the *worst of* model and policy; a policy DECLINE overrides even a favourable model grade
4. **Route** — `AUTO_DECLINE` — RM cannot approve, escalation only
5. **Recourse** — counterfactual engine returns actionable levers (e.g. "reduce requested EMI by Rs. X to bring FOIR under 55%"); applicant-facing adverse-action letter cites the specific reason code, not a black-box score

Why this matters

Demonstrates that **policy always wins** over the model when they disagree — and that every decline ships with a specific, actionable, regulator-defensible reason.

Governance, Auditability & Explainability

Traceability

- Hash-chained `rm_case_events` — every M/H/O step is content-hashed and linked to the previous event
- `content_hash` on the Machine Recommendation itself — tamper-evident, reproducible from inputs
- Model version, policy version (`credit-policy-v1.0`), and report ID all stamped on every decision

Explainability

- SHAP-based per-feature attribution on every score
- Counterfactual recourse — specific, actionable levers for declined or borderline applicants
- Peer benchmarking — segment-scoped comparison against approved borrowers in the same exposure class

Model does not decide alone

The ML model estimates risk; `policy_engine.py` enforces eligibility; `decision_orchestrator.py` determines who may act; a human RM or committee makes the final call. No exposure is auto-approved by the model in isolation.

Recent phases

- Real transaction-derived behavioural features (EMI-miss ratio, income volatility)
- 5-fold CV AUC alongside single-split AUC in the training pipeline
- ALM engine, loan booking & NPA resolution workflow added
- PD models split by Basel exposure class — retired the single unsegmented model
- SHAP-based explainability layer integrated into assessment output

Roadmap

- **Phase 6** — GCP App Engine deployment (in progress)
- **Phase 7** — Operational Risk (TSA / AMA)
- **Phase 8** — Market Risk (VaR)

Thank You

Questions & Discussion

Solution:	Banking Credit Risk Calculator
Coverage:	Credit Risk · Operations · Regulatory · RM · Financials · Reference Data
Deployment:	GCP App Engine

Basel III Compliant, ML-Powered Credit Risk Platform